IGNIS

Integrated Ground-Networked Intelligent UAS System

Full System Architecture Documentation

NASA SBIR Phase I — Nontraditional Aviation Operations for Wildfire Response

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CLASSIFICATION: UNCONTROLLED // FOR OFFICIAL USE

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1. Executive Summary

IGNIS (Integrated Ground-Networked Intelligent UAS System) is an end-to-end drone management and fire prediction platform designed for wildland fire operations. It integrates three major capabilities into a unified, field-deployable system:

- An adaptive fire prediction engine (ELMFIRE ensemble + Gaussian Process uncertainty quantification) that continuously updates probabilistic fire spread forecasts.
- A drone fleet management layer that optimally tasks a limited number of UAS to resolve fire model uncertainty in the locations that most affect predicted fire behavior.
- A communications mesh architecture that simultaneously provides sensor coverage, airborne relay connectivity, and real-time situational awareness to ground crews.

The system is designed to operate in the austere connectivity environments characteristic of wildland fire incidents — remote terrain, no cellular infrastructure, intermittent satellite access — while remaining fully compliant with NWCG PMS 515 standards and FAA regulatory requirements for BVLOS UAS operations.

Key Performance Targets	
Prediction cycle time:	< 20 minutes end-to-end
Telemetry delivery rate:	> 95% (vs. 52% in PAMS TCL1 baseline)
TFR/OI latency:	< 30 seconds (vs. up to 12 minutes in baseline)
Ground crew alert latency:	< 60 seconds for safety-critical events
Drone utilization:	Dual-role (sensing + relay) per platform
Offline operational:	Full capability with zero internet dependency

2. System Overview

2.1 Problem Statement

Wildland fire aerial operations face three compounding challenges that no current system addresses holistically:

1. Fire prediction models are data-starved. RAWS weather stations are spaced 30–50 km apart. Satellite perimeter imagery updates every 4 hours. Between updates, fire managers rely on stale data and expert judgment to position resources and predict fire behavior.
2. Drone operations are manually coordinated. The ATGS manages airspace from a manned aircraft using voice radio and visual observation. This approach cannot scale to support multiple simultaneous BVLOS drone operations, cannot function at night, and cannot accommodate the data volumes produced by modern sensor payloads.
3. Ground crews are informationally isolated. Firefighters on the fireline have no access to the digital common operating picture available to air operations. They receive information via radio relay chains that may be several hops deep, with no guarantee of timeliness.

2.2 Solution Architecture

IGNIS addresses all three problems with a closed-loop architecture where each component feeds the next:

```
┌──────────────────────────────────────────────────────────────────────┐
│                     GROUND STATION (PAMS Case)                       │
│                                                                      │
│  ┌──────────────┐   ┌──────────────┐   ┌──────────────────────┐      │
│  │   ELMFIRE    │   │  GP / EnKF   │   │  IGNIS Orchestrator  │      │
│  │  Ensemble    │◄──│ Assimilation │◄──│  (20-min cycle)      │      │
│  └──────┬───────┘   └──────────────┘   └──────────┬───────────┘      │
│         │ fire spread                             │ mission queue    │
│  ┌──────▼──────────────────────────────────────┐  │  ┌────────────┐  │
│  │          Information Field + Target Selector│  ▼  │            │  │
│  └──────────────────────────────────┬──────────────────────────────┘  │
│                                     │ OI requests                    │
│  ┌──────────────────────────────────▼───────────────────────────────┐│
│  │                    UTM / WFSS Layer                              ││
│  │  Conflict Detection | OI Lifecycle | Conformance | Auth          ││
│  └──────────────────────────────────┬───────────────────────────────┘│
└─────────────────────────────────────┼────────────────────────────────┘
                                      │ cleared flight plans
        ┌─────────────────────────────▼──────────────────────────┐
        │                  DRONE MESH NETWORK                     │
        │  Type 1/2: High-altitude relay + strategic mapping      │
        │  Type 3/4: Low-altitude sensing + ground crew comms     │
        └─────────┬──────────────────────────────┬────────────────┘
                  │ observations                 │ relay
        ┌─────────▼─────────┐           ┌────────▼─────────────┐
        │  Back to IGNIS    │           │  GROUND CREW DEVICES │
        │  (assimilation)   │           │  Fire perimeter + wind│
        └───────────────────┘           │  Escape routes + alerts│
                                        └──────────────────────┘
```

2.3 Operational Concept

A single IGNIS deployment consists of a ground station (ruggedized mini-PC in a PAMS-style wheeled case), a drone fleet of mixed types, and handheld devices for ground crews. The system is fully self-contained — no internet connection is required for core operations. Optional cloud connections (HRRR weather, D-Wave QPU) enhance performance when available.

The system operates in continuous 20-minute cycles. Within each cycle, ELMFIRE runs an ensemble of N=50 fire spread simulations with perturbed inputs drawn from the Gaussian Process uncertainty model. The ensemble output is analyzed to identify which unobserved locations, if measured, would most reduce prediction uncertainty. Drones are tasked to those locations. When drones return observations, the GP and ensemble are updated, and the cycle repeats.

3. NWCG PMS 515 Compliance Framework

All drone operations within IGNIS are governed by NWCG Standards for Fire Unmanned Aircraft Systems Operations, PMS 515, in conjunction with FAA 14 CFR Part 107 and applicable DOI/USFS operational memoranda. Compliance is enforced programmatically — no flight is authorized without the full approval chain being satisfied digitally.

3.1 UAS Fleet Typing

IGNIS operates a mixed fleet classified per PMS 515 Table 1:

Type	Role in IGNIS	Altitude Band	Endurance	Sensors
Type 1 (Fixed-Wing)	Primary strategic relay + perimeter mapping	3,500–8,000 ft AGL	6–14 hrs	EO/Mid-Wave IR, Mode C transponder
Type 2 (Fixed-Wing)	Secondary relay + tactical mapping	3,500–6,000 ft AGL	1–6 hrs	EO/Long-Wave IR, Mode C transponder
Type 3 (Rotorcraft)	Targeted sensing per IGNIS queue	≤ 2,000 ft AGL	20–60 min	Multispectral (FMC), anemometer, EO/IR
Type 4 (Rotorcraft)	Ground crew communications bridge	≤ 1,200 ft AGL	≤ 20 min	900 MHz radio relay, EO video

3.2 Authorization Chain

The following authorization sequence is enforced as a hard gate in the UTM layer. No Operational Intent (OI) may transition to Activated state without each step being digitally confirmed:

1. IC or designee approval obtained and logged in the system.
2. Airspace authorization confirmed: FAA Part 107, SGI Waiver (BVLOS in TFR), DOI/FAA MOA (night or below 1,200 ft AGL), COA (EVLOS), or USFS/FAA MOA as applicable.
3. NOTAM filed with FAA. System generates NOTAM-compatible text automatically from TFR polygon and time window.
4. TFR deconfliction confirmed with dispatch or TFR-controlling authority.
5. Clearance obtained from ATGS, ASM, HLCO, or Lead Plane. One-tap digital approval via ground station UI replaces voice-only coordination.
6. OI submitted to WFSS and validated against active TFR volume and existing OIs. Strategic conflict detection run automatically.

7. OI transitions to Accepted. Drone authorized to launch.

3.3 Call Sign Management

IGNIS automatically assigns and tracks NWCG-standard call signs for all managed aircraft. Call signs follow the format: [Unmanned] [R/F][Type][Number]. Examples: Unmanned R31, Unmanned F12, Unmanned R42. The system broadcasts call signs on both assigned Victor (AM) and air-to-ground (FM) frequencies. For BVLOS operations where no human operator is monitoring audio, the system generates automated blind radio calls at configurable intervals.

3.4 Separation and Deconfliction Requirements

Per PMS 515, UAS must give way to all manned aircraft at all times. IGNIS enforces this through three layers:

- Altitude band enforcement: Type 3 drones cannot be authorized above 2,500 ft AGL regardless of IGNIS mission requests. Hard ceiling enforced in OI validation.
- ADS-B proximity alerting: When an ADS-B target enters the TFR or approaches within 500 ft horizontal / 200 ft vertical of any active UAS altitude band, all drones in that band execute a pre-programmed hold maneuver and the ATGS is alerted.
- Lost-link protocol: Every OI must specify a lost-link behavior before authorization is granted — loiter, return-to-LRZ, or controlled descent to a designated safe zone. This is pre-programmed into the flight controller before launch.

3.5 Ground Crew Safety Requirements

Per PMS 515 BVLOS guidelines, all ground resources must be briefed about UAS operations above them. IGNIS addresses this through:

- Ground crew devices display a persistent 'UAS Overhead' indicator showing drone positions within 500 ft horizontal of crew locations.
- When any drone descends below 500 ft AGL within 300 m of a crew position, an audible and visual alert is issued on all crew devices.
- Crew GPS positions are visible to the ATGS and all drone operators in the common operating picture, ensuring spatial awareness of human assets at all times.

4. IGNIS Core Prediction Engine

The prediction engine runs continuously on the ground station, completing a full cycle every 20 minutes. It consists of six tightly coupled components: Terrain Manager, Gaussian Process Prior, ELMFIRE Ensemble Engine, Information Field Computation, Target Selector, and Data Assimilation.

4.1 Terrain Manager

Static data loaded once at deployment from LANDFIRE via the landfire-python library. Provides elevation, slope, aspect, and Anderson 13 fuel models at 50m resolution for the incident domain. Fuel model parameters (load, SAV ratio, bed depth, moisture of extinction, heat content) are stored as published constants from Andrews 2018, RMRS-GTR-371. The terrain data is immutable after initialization — it does not change during a cycle.

```
TerrainData:
  elevation:   float32[rows, cols]    # meters MSL
  slope:       float32[rows, cols]    # degrees
  aspect:      float32[rows, cols]    # degrees from north
  fuel_model:  int8[rows, cols]       # Anderson 13 IDs (1-13)
  resolution:  50.0 m
  origin:      (lat, lon)             # NW corner of domain
```

4.2 Gaussian Process Prior

The GP maintains a spatially resolved estimate of fuel moisture content (FMC) and wind fields with calibrated uncertainty at every grid cell, conditioned on all available observations. Observations include RAWS station data (~50 km spacing), HRRR weather forecast (pulled once per cycle when connectivity is available, cached otherwise), and all previous drone measurements.

The kernel uses terrain-aware distance — two points on the same aspect and elevation band are more correlated than two points at equal Euclidean distance on opposite sides of a ridge. Correlation lengths are set to physically motivated defaults: approximately 1–2 km for FMC and 5 km for wind, fitted from RAWS data when sufficient observations are available.

The critical property exploited by the target selector is the closed-form conditional variance update: when a new observation is added at location x_new, the posterior variance at every other grid cell can be updated in a single vectorized NumPy operation with no matrix inversion. This makes the greedy selector computationally tractable even for large grids.

```
GP Posterior Update (per new observation at x_new):

  sigma2_updated(x) = sigma2(x) - k(x, x_new)^2 / (k(x_new, x_new) + sigma2_noise)

  Cost: one vectorized operation over all D grid cells
  No matrix inversion, no GP refitting
```

```
Called K times per cycle (once per drone placement)
```

4.3 ELMFIRE Ensemble Engine

ELMFIRE (Eulerian Level-set Model of FIRE spread) replaces the custom Rothermel cellular automaton described in initial IGNIS design. ELMFIRE is a GPU-native, physics-based fire spread model developed at USFS, using the same Rothermel rate-of-spread physics but solving them as a level-set PDE rather than cellular automata. This eliminates grid-direction bias, improves fire front geometry accuracy, and provides better handling of backing fires.

ELMFIRE ensemble execution: N=50 members run with perturbed FMC and wind fields drawn from the GP posterior. Perturbations are spatially correlated — cells near RAWS stations (low GP variance) receive small perturbations while cells in data-sparse terrain (high GP variance) receive large perturbations. The correlation structure is generated via circulant embedding with FFT, cost O(D log D), avoiding the O(D^3) full covariance matrix.

Parameter	Value	Notes
Ensemble size (N)	50 members	Sufficient for stable variance estimates; increase to 100 for high-complexity terrain
Domain resolution	50 m	Resampled from 30m LANDFIRE for computational tractability
Forecast horizon	60 minutes	Covers 2–3 prediction cycles ahead
GPU memory required	~12 GB	For N=50 on 200x200 domain; RTX 4090 (24 GB) provides headroom
Single-member runtime	30–60 sec	GPU-accelerated ELMFIRE on RTX 4090
Full ensemble (batched)	< 5 minutes	With sufficient GPU VRAM to batch members

4.4 Information Field Computation

The information field w(x) quantifies, at every grid cell, the value of a drone measurement at that location. It combines three factors: GP variance (how uncertain is this location), sensitivity (does uncertainty here actually affect the predicted fire trajectory), and observability (can the drone sensor reliably measure this variable at this location).

```
Information Field Computation:

Step 1 — Sensitivity (per variable v, per cell c):
  S_v(c) = corr(arrival_times[:, c], perturbation_fields[:, c])
  Computed via element-wise correlation across N ensemble members
  Cost: one matrix operation, milliseconds total

Step 2 — Observability (sensor accuracy by location):
  D_fmc(x)  = 0.86 baseline, degraded near active fire by smoke proxy
  D_wind(x) = 0.90 baseline, degraded by terrain complexity
```

```
Step 3 — Information value (combined):
  w(x) = GP_var_fmc(x) * |S_fmc(x)| * D_fmc(x)
        + GP_var_wind(x) * |S_wind(x)| * D_wind(x)
        + GP_var_winddir(x) * |S_winddir(x)| * D_wd(x)

Step 4 — Overlays:
  w(x) *= priority_weight(x)   # operator-defined priority regions
  w(x)  = 0  if x in exclusion_zone
```

4.5 Target Selector

Two selection algorithms run in parallel each cycle, with their outputs compared for quality on the same counterfactual evaluation framework:

4.5.1 Greedy Selector

Iteratively selects the highest-value cell, then updates the GP variance to account for the information that observation would provide, naturally handling spatial redundancy through the GP kernel's correlation structure. Achieves a provable (1 - 1/e) ≈ 63% approximation ratio to the optimal solution under Gaussian assumptions (Krause et al., 2008). Runtime scales as O(K * D) where K is the number of drones and D is the grid size — milliseconds for typical domains.

4.5.2 QUBO Selector

Encodes the same information-gain-with-redundancy problem as a Quadratic Unconstrained Binary Optimization matrix. Linear terms encode individual location value (w_i). Quadratic terms encode spatial redundancy via ensemble covariance. Cardinality penalty enforces exactly K selections. Solved via fallback chain: D-Wave QPU (if satellite connectivity available) → simulated annealing (always available on ground station GPU) → greedy (always succeeds). Both selectors produce the same output format and are compared on equal footing each cycle.

4.6 Data Assimilation

Two parallel update mechanisms run after each drone observation batch:

- GP update: New observations are added to the conditioning set. Posterior variance drops near observed locations, reducing perturbation magnitude in those regions for the next ensemble run.
- Ensemble Kalman Filter (EnKF): Adjusts each ensemble member's state to be consistent with drone observations. Gaspari-Cohn localization tapering limits spurious long-range correlations, preventing observations from incorrectly updating distant unobserved regions.

Replan triggers are computed after assimilation: if total posterior variance drops by more than 20%, the cycle is flagged as high-information. If observed wind deviates from the prior mean by more than 30 degrees at any location, an immediate partial re-solve is triggered without waiting for the full 20-minute cycle.

5. UTM and Airspace Management Layer

The UTM layer is derived from NASA's FREDDIE (Federated Airspace Management Framework) and the PAMS/WFSS architecture developed under the ACERO project. It handles the full OI lifecycle, strategic conflict detection, conformance monitoring, and the interface between IGNIS mission requests and regulatory compliance.

5.1 Wildland Fire Service Supplier (WFSS)

Each ground station runs a WFSS instance that provides the following capabilities:

- Constraint Sharing: TFR volumes ingested and broadcast to all PAMS cases in the network. Periodic rebroadcast at 60-second intervals ensures eventual delivery despite intermittent connectivity.
- Operational Intent (OI) Management: Full OI lifecycle per ASTM F3548-21, adapted for wildfire operations. See Section 5.2 for state diagram.
- Strategic Conflict Detection: OIs compared against TFR volume and all other active OIs. Conflicts flagged with nature of overlap (spatial volume and time window). Users prompted to replan before activation.
- Conformance Monitoring: Active drones tracked against their filed OI volumes. Alerts issued and broadcast to all WFSS instances when a drone exits its authorized volume.
- Data Archiving: All AMF messages (TFR, OI, telemetry) logged with timestamps for post-incident analysis and regulatory compliance.

5.2 Operational Intent State Machine

The IGNIS OI lifecycle adapts the ASTM F3548-21 standard with two wildfire-specific modifications: (1) the Contingent state is replaced by a pre-declared contingency volume extension that activates automatically on nonconformance, and (2) OI approval is enforced digitally through the ATGS interface rather than voice radio.

```
OI State Diagram:

  [IGNIS generates mission] → DRAFT
       │
       ▼ (WFSS validates against TFR + existing OIs)
    ACCEPTED
       │
       ▼ (ATGS one-tap digital approval)
    ACTIVATED  ──────────────────────────────────────────┐
       │                                                  │
       ▼ (drone exits OI volume or time window)          │
  NONCONFORMING  ─── (drone returns to volume) ──────────┘
       │                                                  │
       ▼ (timeout or operator action)                     │
    CONTINGENT VOLUME ACTIVE  ──── (resolves) ────────────┘
       │
```

```
       ▼ (drone landed, telemetry stops)
     ENDED

   Key difference from UTM: Contingent state pre-declared as
   volume expansion (+200m lateral, +500ft vertical) around
   nominal OI. Activates automatically, no operator action needed.
   Eliminates lost-link gap from original PAMS design.
```

5.3 IGNIS-to-WFSS Bridge

This bridge is the critical integration point between the prediction engine and the airspace management layer. It converts IGNIS MissionRequest objects into WFSS-compliant OI volumes and manages the authorization workflow.

```
MissionRequest (from IGNIS):           OI Volume (for WFSS):
  target: (lat, lon)           →         polygon: 200m radius circle
  information_value: float     →         (priority score for scheduling)
  dominant_variable: str       →         (logged for post-analysis)
  substitutes: [(lat,lon)]     →         (fallback OI volumes if conflict)
  expiry_minutes: float        →         time_window: [now, now+expiry]
  drone_type: UASType          →         altitude_range: per PMS 515 typing

Authorization workflow (automated):
  1. Validate OI against TFR volume
  2. Run strategic conflict detection vs. all active OIs
  3. If no conflicts → push to ATGS approval queue
  4. ATGS approves via one-tap UI (replaces voice radio)
  5. OI transitions to Accepted → drone assigned and briefed
  6. On launch confirmation → OI transitions to Activated
```

5.4 Airspace Conflict Detection

Conflict detection runs in four dimensions — spatial volume (x, y, z) and time — and at three levels of severity:

Conflict Type	Detection Method	Response
TFR boundary violation	OI volume intersects TFR exclusion zone	OI rejected; IGNIS generates substitute target
OI-OI spatial conflict	4D volume overlap between two active OIs	Warning issued; users prompted to replan
Altitude band violation	OI altitude exceeds PMS 515 ceiling for UAS type	Hard block; OI cannot be activated
ADS-B proximity alert	ADS-B target within 500 ft horizontal / 200 ft vertical of active UAS	Immediate hold issued to all drones in band; ATGS alerted
Lost-link / conformance breach	Telemetry stops or drone exits OI volume	Contingency volume activates; alert broadcast to all WFSS instances

6. Drone Fleet Management

The fleet management layer bridges the UTM authorization system and physical drone operations. It handles role allocation, coverage optimization, path planning, and real-time reallocation in response to IGNIS replan triggers.

6.1 Dual-Role Architecture

Each drone in the IGNIS fleet serves two simultaneous roles: a primary mission role (sensing or relay) and a secondary mesh networking role. This is the key architectural departure from PAMS, where a dedicated relay drone was required as a separate asset. In IGNIS, every drone is a mesh node, so adding drones for improved fire coverage automatically improves network redundancy.

Why This Fixes the PAMS Reliability Problem
PAMS TCL1 result: 52% telemetry success rate with 1 relay drone + 3 ground nodes
Root cause: Single relay node = single point of failure for entire mesh
IGNIS approach: Every drone is a relay node; N drones = N-1 relay redundancy Expected result: Network failure requires ALL high-altitude drones to fail simultaneously
Scales naturally: more drones = better coverage = better comms

6.2 Fleet Role Allocation

Given a fleet of N drones, the allocation algorithm balances relay topology coverage against IGNIS sensing requirements. The relay fraction is a function of terrain complexity and the number of ground nodes that need connectivity:

```
Fleet Allocation Algorithm:

  relay_fraction = f(terrain_complexity, ground_node_count, K_sensing)

  Defaults (adjustable by ATGS):
    N=3:  1 relay (Type 1/2) + 2 sensing (Type 3)
    N=5:  2 relay (Type 1/2) + 3 sensing (Type 3)
    N=8:  2 relay (Type 1/2) + 4 sensing (Type 3) + 2 ground-crew comms (Type 4)
    N=10: 3 relay + 5 sensing + 2 ground-crew comms

  Relay drones: Fixed waypoints optimized for LOS to all ground nodes
  Sensing drones: Dynamic waypoints from IGNIS mission queue
  Ground-crew comms drones (Type 4): Orbit above crew positions at 800-1200 ft AGL
```

6.3 Coverage Optimization

Relay drone positions are computed by solving a facility location problem over the terrain: find the set of R airborne positions that maximizes line-of-sight connectivity to all ground nodes while maintaining mutual LOS between relay drones. The terrain digital elevation model from LANDFIRE is used to compute LOS profiles.

```
Relay Position Optimization:

  Inputs:
    - Ground node positions (ground station + ground crew devices)
    - Terrain DEM at 50m resolution
    - R = number of relay drones available
    - Altitude constraint: relay drones operate at 3,500-8,000 ft AGL

  Objective:
    Minimize max_hop_count(any_node, ground_station)
    Subject to: LOS(relay_i, relay_j) for adjacent relays
                LOS(relay_i, ground_node_j) for assigned nodes
                altitude constraints per PMS 515

  Method: Greedy facility location (same submodularity argument as IGNIS selector)
  Runtime: < 1 second for R <= 4, N_nodes <= 20
```

6.4 Dynamic Reallocation

When IGNIS identifies a high-priority replan trigger (wind shift > 30 degrees or new high-uncertainty region outside current drone coverage), the fleet manager can interrupt a low-priority sensing mission and redirect a drone within a single 20-minute cycle. The reallocation sequence is:

8. IGNIS flags replan trigger with new target set.
9. Fleet manager identifies lowest-priority active sensing mission.
10. New OI submitted for redirected drone, validated by WFSS.
11. ATGS one-tap approval of new OI.
12. Existing OI transitioned to Ended (drone rerouted mid-flight if needed).
13. New OI activated, drone proceeds to new waypoints.

Total reallocation latency target: under 3 minutes from trigger to new OI activation.

6.5 Path Planning

Path planning converts IGNIS-selected target locations into feasible drone flight plans that respect OI volume constraints, altitude band limits, and PMS 515 separation requirements. For the initial deployment, nearest-neighbor routing from a staged launch and recovery zone (LRZ) is used. Future versions will incorporate minimum-time routing with obstacle avoidance for complex terrain.

Path-integrated observations: drones do not only measure at waypoints — they observe every grid cell they fly over. The path planner computes all cells within the sensor footprint along each flight leg using Bresenham's line algorithm with a 3-cell-wide camera footprint at nominal flight altitude. These additional observations are included in the assimilation step, making targeted flights more efficient than the path alone would suggest.

7. Ground Crew Communications System

The ground crew communications subsystem extends the IGNIS common operating picture to firefighters on the fireline — the user population most at risk and least served by existing digital situational awareness tools. It is designed for extreme simplicity of use: no training required beyond a 5-minute briefing, single-button status reporting, and battery life exceeding a full operational period (10+ hours).

7.1 Device Architecture

Each ground crew supervisor carries a ruggedized Android tablet or smartphone with the IGNIS ground crew application. The device communicates with the nearest Type 4 drone overhead via 900 MHz license-free radio link (900 MHz chosen for superior range and penetration through smoke relative to 2.4/5.8 GHz). The Type 4 drone relays traffic to the WFSS ground station via the mesh network.

```
Ground Crew Communication Path:

  Ground crew device
    └── 900 MHz radio link (range: 3-5 km line-of-sight)
          └── Type 4 drone (orbiting at 800-1,200 ft AGL)
                └── Silvus/DoodleLabs mesh (S-band)
                      └── Type 1/2 relay drone (3,500-8,000 ft AGL)
                            └── Silvus/DoodleLabs mesh
                                  └── WFSS ground station
                                        └── IGNIS + common operating picture

  Latency budget: < 2 seconds end-to-end for non-safety messages
  Safety alerts:  Priority interrupt, < 500 ms, retransmitted every 5 sec
                  until acknowledged
```

7.2 Information Downlink to Ground Crews

The following data is pushed to ground crew devices on each IGNIS cycle completion (approximately every 20 minutes) and on any safety-critical event trigger:

Data Element	Source	Update Frequency
Current fire perimeter (GeoJSON)	ELMFIRE ensemble mean	Every 20-min cycle
20-min predicted perimeter + uncertainty band	ELMFIRE ensemble spread	Every 20-min cycle
Wind speed and direction at crew position	GP posterior at crew GPS	Every 20-min cycle

Data Element	Source	Update Frequency
All crew GPS positions	Uplink from all devices	Continuous (30-sec poll)
Active aircraft overhead	WFSS ADS-B + OI registry	Continuous
Escape routes and safety zones	Pre-designated + IGNIS updated	On change
Hazard alerts (wind shift, blowup)	IGNIS replan trigger	Immediate, priority interrupt
UAS overhead indicator	WFSS conformance monitor	Continuous

7.3 Information Uplink from Ground Crews

Ground crew observations feed back into IGNIS as soft data points. The ground crew app provides three reporting mechanisms, designed for use with gloves and under stress:

- One-tap status: OK / Needs Support / Emergency. Status visible to IC and ATGS immediately.
- Fire observation: Pre-set options for spotting distance, flame length estimate, rate-of-spread qualitative (slow / moderate / running). Translated to IGNIS soft constraints — e.g., 'running uphill NE' updates the wind direction prior in that sector.
- Free text (optional): For unusual observations. Logged but not automatically assimilated.

7.4 Safety Alert Protocol

Safety alerts bypass normal message scheduling and are transmitted as priority interrupts through every available path simultaneously (mesh network, direct 900 MHz, and satellite fallback if available). Alert conditions and their triggers:

Alert Type	Trigger Condition	Response Required
Wind shift warning	IGNIS detects > 30-degree wind direction change at any observed location	Acknowledge within 60 sec; auto-escalate to ATGS if not acknowledged
Blowup potential	ELMFIRE ensemble shows > 30% probability of area ignition in next 20 min	Immediate acknowledgment; evacuation route highlighted
UAS approach	Any drone within 500 ft horizontal and 500 ft AGL of crew position	Advisory only; no action required
Communication degraded	No uplink from crew device for > 5 minutes	ATGS alerted; last known position displayed on COP
Crew emergency	Emergency button pressed on crew device	Immediate broadcast to all devices; ATGS and IC alerted; GPS position pinned

8. Network Architecture

The IGNIS network is designed for resilience-first operation in environments with no cellular or internet infrastructure. It uses a multi-layer mesh architecture where every airborne platform is a network node, and no single failure can partition the network.

8.1 Network Layers

Layer	Technology	Role	Fallback
Primary airborne mesh	Silvus Streamcaster 4200 E+ (S-band)	High-bandwidth data between drones and ground station	DoodleLabs mesh
Secondary airborne mesh	DoodleLabs (2.4/5.8 GHz)	Backup inter-drone and drone-to-ground link	Store-and-forward DTN
Ground crew link	900 MHz license-free radio	Device-to-Type-4-drone last-mile	Direct SMS-style text if mesh down
Satellite backhaul	Starlink / Iridium (optional)	HRRR weather pull, D-Wave QPU, remote COP access	None (graceful degradation)
Local ground mesh	Ethernet / WiFi	PAMS case to radio node connection	USB tether

8.2 Message Priority and Store-and-Forward

The PAMS TCL1 evaluation showed 52% telemetry delivery due to no retransmit logic for telemetry packets. IGNIS implements a four-tier priority queue with store-and-forward for all message types:

Priority	Message Type	Delivery Guarantee	Max Latency
P0 — Emergency	Safety alerts, crew emergency, lost-link	Retransmit every 5 sec until ACK; broadcast all paths	< 500 ms
P1 — Safety	Wind shift alerts, blowup potential, ADS-B proximity	Retransmit every 15 sec until ACK	< 5 sec
P2 — Operational	OI state transitions, TFR updates, conformance alerts	Periodic rebroadcast at 60-sec intervals	< 60 sec
P3 — Telemetry	Drone position, velocity, sensor readings	Delta-compressed, adaptive rate; store-and-forward buffer	< 5 min acceptable

Priority	Message Type	Delivery Guarantee	Max Latency
P4 — Background	Fire perimeter updates, map tiles, IGNIS cycle reports	Best-effort, dropped if congested	As available

8.3 Telemetry Compression

Raw telemetry at full rate was the primary cause of the 52% delivery failure in PAMS. IGNIS uses three compression strategies applied in combination:

- Trajectory state compression: transmit position + velocity vector + timestamp. Receivers dead-reckon position between updates. Only deviations from dead-reckoned position trigger a correction packet.
- Delta encoding against OI: when a drone is conforming (flying within its filed OI volume), only transmit deviation from the filed trajectory. Conforming flight requires minimal bandwidth.
- Adaptive rate control: telemetry rate is a function of link quality (RSSI/SNR logged per radio pair). Full rate at strong signal, reduced rate as link degrades, minimum heartbeat at poor link. Rate restored immediately when link improves.

8.4 Delay-Tolerant Networking

For scenarios where connectivity is intermittent rather than degraded, IGNIS implements Bundle Protocol (RFC 9171) for store-and-forward operation. OI state transitions and safety alerts are bundled and stored locally if the destination is unreachable, then delivered when connectivity resumes. This ensures that even in the worst-case connectivity scenario — a drone flying behind a ridge for 10 minutes — all state transitions are eventually delivered and the common operating picture converges.

9. Hardware and Deployment

9.1 Ground Station

The ground station runs all software components: IGNIS (ELMFIRE ensemble, GP, information field, target selection, assimilation), WFSS (UTM/airspace management), fleet management, and the common operating picture display. It is packaged in a PAMS-style ruggedized wheeled case for field deployment.

Component	Specification	Purpose
Compute	Intel Core i9 / AMD Ryzen 9, 32 GB RAM	IGNIS orchestrator, GP, EnKF, fleet management
GPU	NVIDIA RTX 4090 (24 GB VRAM)	ELMFIRE ensemble (N=50 batched), QUBO simulated annealing
Storage	2 TB NVMe SSD	LANDFIRE terrain cache, mission data archive, telemetry log
Radio interface	Silvus SC4200E+ (primary), DoodleLabs (backup)	Mesh network uplink
ADS-B receiver	uAvionix pingStation or equivalent	Manned aircraft situational awareness
Power	AC mains + 200 Wh LiFePO4 battery	2-4 hrs autonomous operation; vehicle 12V charging
Display	15" ruggedized touchscreen	ATGS common operating picture interface
Case	Pelican 1650 or equivalent	Field portability; IP67 dust/water resistance

9.2 Drone Platform Requirements

IGNIS is platform-agnostic at the airframe level. The following sensor and communication requirements must be met for each drone type:

Requirement	Type 1/2 (Relay)	Type 3 (Sensing)	Type 4 (Crew Comms)
Mode C transponder	Required (PMS 515)	Not required	Not required
ADS-B out	Required (Type 1/2)	Not required	Not required
Silvus/DoodleLabs radio	Required	Required	Optional
900 MHz radio relay	Not required	Not required	Required

Requirement	Type 1/2 (Relay)	Type 3 (Sensing)	Type 4 (Crew Comms)
Multispectral camera	Not required	Required (FMC measurement)	Not required
Anemometer	Not required	Recommended	Not required
Thermal/IR camera	Optional	Recommended	Not required
GPS accuracy	< 3 m CEP	< 3 m CEP	< 10 m CEP
Telemetry downlink	MAVLink or equivalent	MAVLink or equivalent	MAVLink or equivalent

9.3 Ground Crew Devices

Ground crew devices must be ruggedized, simple, and battery-capable of a full operational period. Minimum requirements: Android 10 or later, IP67 dust/water resistance, 10+ hour battery life, built-in 900 MHz radio capability (or external dongle), screen readable in direct sunlight (> 600 nit brightness). Recommended devices include Getac ZX10, Samsung Galaxy XCover Pro, or Kyocera DuraForce Ultra 5G.

10. Software Architecture

10.1 Repository Structure

```
ignis/
├── core/
│   ├── types.py              # Shared dataclasses (TerrainData, GPPrior, etc.)
│   ├── config.py             # Constants, fuel params, PMS 515 altitude limits
│   └── utils.py              # UTM projection, distances, geometry
├── terrain/
│   └── terrain.py            # LANDFIRE loader, DEM processing
├── prediction/
│   ├── gp.py                 # GP prior, conditional variance update
│   ├── elmfire_engine.py     # ELMFIRE ensemble wrapper
│   ├── information.py        # Sensitivity + information field
│   └── assimilation.py       # EnKF + GP observation update
├── selectors/
│   ├── greedy.py             # Greedy selector with GP variance update
│   ├── qubo.py               # QUBO construction + solver fallback chain
│   └── baselines.py          # Uniform grid, fire-front following
├── utm/
│   ├── wfss.py               # WFSS OI lifecycle, conflict detection
│   ├── bridge.py             # IGNIS MissionRequest → WFSS OI conversion
│   ├── airspace.py           # ADS-B integration, altitude enforcement
│   └── auth_chain.py         # PMS 515 authorization gate
├── fleet/
│   ├── allocator.py          # Role allocation, relay position optimization
│   ├── path_planner.py       # Waypoint generation, path-integrated obs
│   └── reallocator.py        # Dynamic reallocation on replan triggers
├── network/
│   ├── mesh.py               # Silvus/DoodleLabs radio interface
│   ├── priority_queue.py     # P0-P4 message priority + store-and-forward
│   ├── telemetry.py          # Delta compression, adaptive rate control
│   └── dtn.py                # Bundle Protocol RFC 9171 implementation
├── ground_comms/
│   ├── crew_feed.py          # Downlink data package assembly
│   ├── crew_uplink.py        # Observation ingestion + IGNIS soft constraint
│   └── alert_manager.py      # Safety alert priority interrupt
├── orchestrator.py           # 20-minute cycle coordinator
├── evaluation.py             # Counterfactual comparison framework
└── visualization.py          # Common operating picture display

apps/
├── ground_station/           # ATGS desktop interface (Python/Qt or Electron)
└── crew_app/                 # Ground crew Android application

scripts/
├── run_cycle.py              # Single cycle execution
├── run_comparison.py         # Multi-strategy comparison
└── sim_scenario.py           # Simulation-only mode for testing
```

10.2 Key Interfaces

```
# IGNIS → UTM Bridge
class MissionRequest:
    target: tuple[float, float]        # (lat, lon)
    information_value: float           # w_i from information field
    dominant_variable: str             # 'fmc' | 'wind_speed' | 'wind_dir'
    substitutes: list[tuple]           # fallback locations if OI conflicts
    expiry_minutes: float              # OI time window duration
    drone_type: UASType                # determines altitude band
    priority: int                      # 0=emergency replan, 1=normal, 2=low

# UTM → IGNIS Feedback
def ingest_observation(obs: DroneObservation) -> Optional[MissionQueue]: ...
def add_exclusion_zone(polygon, reason) -> None: ...
def add_priority_region(polygon, weight) -> None: ...

# Ground Crew → IGNIS
def ingest_crew_observation(obs: CrewObservation) -> None:
    # Translates qualitative fire observations to GP soft constraints
    # 'running uphill NE' → wind_direction_prior update in NE sector
    # 'spotting 300m SE' → effective spotting distance observation
```

10.3 Degradation Contracts

Every component has a defined failure mode that allows the system to continue operating in a degraded but safe state:

Component	Failure	Degraded Behavior
LANDFIRE API	Unreachable at deployment	Use synthetic terrain (fractal DEM + default fuel model 2)
ELMFIRE	Timeout or crash	Use last valid ensemble; flag predictions as stale after 40 min
GP fitting	Insufficient RAWS data (< 2 stations)	Constant prior variance everywhere; note in COP
QUBO / D-Wave	Unreachable or timeout	Automatic fallback to simulated annealing, then greedy
EnKF	No observations in last cycle	Pass prior through unchanged; GP variance increases
Mesh network	Primary (Silvus) failure	Automatic failover to DoodleLabs backup
Satellite backhaul	Unavailable (normal case)	No HRRR update; cached forecast used; D-Wave unavailable
Ground crew device	No uplink for > 5 min	Last known position displayed; ATGS alerted
ADS-B receiver	Failure	Voice radio monitoring required; alert operator; no automated proximity alerts

11. Operational Deployment and Verification

11.1 Deployment Sequence

IGNIS is designed to be operational within 30 minutes of arriving at an incident:

14. Transport: Ground station in wheeled PAMS case. Drone fleet in vehicle transport cases. Ground crew devices pre-charged and loaded with incident GIS data.
15. Site survey: Identify LRZ (flat, clear of obstacles, away from manned aircraft landing zones). Establish ground station location with clear sky view for satellite and ADS-B.
16. Network initialization: Power on ground station. Connect Silvus radio via ethernet. Verify mesh connectivity between ground station radio node and first relay drone on the ground.
17. Terrain and data load: IGNIS auto-loads LANDFIRE data for incident bounding box (pulled from cache if pre-loaded, or downloaded via satellite if available). Load incident perimeter from current situation report.
18. Authorization chain: Confirm IC approval. Verify airspace authorization type. File NOTAM (auto-generated by IGNIS from TFR polygon). Coordinate with ATGS for clearance.
19. First drone launch: Type 1/2 relay drone launched first, climbs to altitude, establishes mesh relay. System verifies >90% telemetry delivery before launching sensing drones.
20. IGNIS first cycle: Run initial ensemble. Generate first mission queue. ATGS reviews and approves sensing drone OIs. Sensing drones launch.
21. Ground crew brief: Distribute crew devices. 5-minute app briefing. Verify two-way communication with ground station.

11.2 Verification and Validation

The following tests must pass before IGNIS is cleared for operational deployment at a wildland fire incident:

Test	Pass Criterion	NWCG Reference
Authorization chain enforcement	No OI activates without complete digital authorization chain	PMS 515 §3
Altitude band enforcement	Type 3 OI rejected if altitude exceeds 2,500 ft AGL	PMS 515 Table 1
ADS-B proximity alert	Alert issued within 5 sec of simulated ADS-B target entering proximity zone	PMS 515 §7
Lost-link response	Drone executes pre-programmed contingency within 30 sec of simulated link loss	PMS 515 §8
Telemetry delivery rate	> 95% of telemetry packets delivered within 5-minute window	Exceeds PAMS TCL1 baseline

Test	Pass Criterion	NWCG Reference
Safety alert latency	P0 alert delivered to all crew devices within 500 ms in benign conditions	BVLOS safety requirement
Call sign broadcast	Blind AM/FM calls generated correctly for BVLOS operations	PMS 515 §6
Ground crew UAS indicator	UAS overhead indicator updates within 5 sec of drone entering 500 ft proximity	PMS 515 BVLOS §5
Full cycle time	Complete IGNIS cycle (ensemble + targeting + OI generation) < 20 minutes	System performance target
Offline operation	All core functions operational with zero internet connectivity for 4+ hours	Operational requirement

11.3 NASA SBIR Phase I / Phase II Milestones

Milestone	Deliverable	Phase
M1: Architecture validation	System design document (this document); simulation environment operational	Phase I
M2: ELMFIRE integration	ELMFIRE ensemble running on ground station GPU; GP prior validated against RAWS data	Phase I
M3: IGNIS cycle demonstration	Full 20-minute cycle in simulation; greedy vs. QUBO comparison results	Phase I
M4: UTM layer integration	WFSS OI lifecycle; IGNIS-WFSS bridge; authorization chain enforced	Phase I
M5: Network prototype	Silvus mesh + DoodleLabs backup; priority queue; telemetry compression	Phase II
M6: Ground crew app	Android app; 900 MHz link; safety alert protocol validated	Phase II
M7: TCL field evaluation	Operationally relevant environment; comparison against PAMS baseline metrics	Phase II
M8: Multi-cycle assimilation	EnKF + GP update over 5+ cycles; entropy reduction demonstrated	Phase II
M9: Phase III transition	Partnership with CAL FIRE / USFS / commercial UAS operator for operational trial	Phase II exit

12. Risk Register

Risk	Likelihood	Impact	Mitigation
ELMFIRE ensemble does not complete within 20-min cycle on available GPU hardware	Medium	High	Reduce ensemble size to N=25 as fallback; extend cycle to 30 min; upgrade to dual-GPU if needed
D-Wave QPU unavailable (no satellite connectivity)	High (expected)	Low	Simulated annealing on GPU is primary solver; D-Wave is enhancement only
FMC sensor accuracy degrades near active fire due to smoke	High	Medium	Smoke optical depth proxy degrades D_fmc observability; IGNIS redirects drones to non-smoke-affected sectors
GPS accuracy degrades in mountainous canyons	Medium	Medium	Terrain-aided navigation; relative positioning via mesh TDOA; note in proposal as future work
Ground crew device battery fails mid-shift	Medium	Medium	Vehicle charging in LRZO; spare devices; last known position retained in COP for 2 hours
ATGS unfamiliar with digital approval workflow	High (initial)	Medium	One-tap approval requires < 3 seconds; falls back to voice radio if tablet unavailable; training materials
Single relay drone failure partitions mesh network	Low	High	Multi-relay topology; automatic rerouting; DTN store-and-forward bridges partitions
Regulatory changes to BVLOS authorization during project	Low	High	Monitor FAA Reauthorization Act Section 910; architecture is authorization-type-agnostic

13. Glossary

Term	Definition
ADS-B	Automatic Dependent Surveillance-Broadcast. Aircraft position broadcast system.
ATGS	Air Tactical Group Supervisor. Manages airspace over an incident.
BVLOS	Beyond Visual Line of Sight. UAS operations where pilot cannot see the aircraft.
COA	Certificate of Authorization. FAA authorization for UAS operations.
DTN	Delay-Tolerant Networking. Store-and-forward protocol for intermittent connectivity.
ELMFIRE	Eulerian Level-set Model of FIRE spread. GPU-native wildfire spread model (USFS).
EnKF	Ensemble Kalman Filter. Data assimilation method for updating model state.
FMC	Fuel Moisture Content. Key driver of fire spread rate; primary IGNIS sensing target.
FTA	Fire Traffic Area. 5 nm radius airspace established around a wildland fire.
GP	Gaussian Process. Probabilistic model providing spatially continuous uncertainty estimates.
HRRR	High-Resolution Rapid Refresh. NOAA hourly weather forecast model.
IC	Incident Commander. Highest authority at an incident.
IGNIS	Integrated Ground-Networked Intelligent UAS System (this system).
LRZ	Launch and Recovery Zone. Designated UAS takeoff and landing area.
NWCG	National Wildfire Coordinating Group. Sets standards for wildland fire operations.
OI	Operational Intent. Declared airspace volume + time window for a drone operation.
PAMS	Portable Airspace Management System. NASA ACERO prototype airspace management system.
PMS 515	NWCG Standards for Fire Unmanned Aircraft Systems Operations.
QUBO	Quadratic Unconstrained Binary Optimization. Problem formulation for combinatorial optimization.
RAWS	Remote Automated Weather Station. Sparse ground weather network.
SGI	Special Government Interest waiver. FAA authorization for BVLOS in a TFR.

Term	Definition
TFR	Temporary Flight Restriction. FAA-established restricted airspace around an incident.
UTM	UAS Traffic Management. NASA/FAA framework for low-altitude drone operations.
WFSS	Wildland Fire Service Supplier. PAMS adaptation of the UTM USS concept.